Question 15 1 mark

Given $f(x) = -2x - 5$ and $g(x) = x + 7$, identify the equation of $h(x) = f(g(x))$.

a) $h(x) = -2x^2 - 35$

b) $h(x) = -2x - 19$

c) $h(x) = -2x^2 - 19x - 35$

d) $h(x) = -2x + 2$

Question 16 1 mark

The function $y = f(x)$ has a domain of $[2, 5]$ and a range of $[-6, 3]$. Identify the range of the function $y = f^{-1}(x)$.

a) $[-6, 3]$

b) $[-5, 2]$

c) $[-3, 6]$

d) $[2, 5]$

Question 17 1 mark

Identify the function that has a graph with a point of discontinuity (hole) at $x = -3$.

a) $y = \dfrac{x+3}{x^2-9}$

b) $y = \dfrac{x-3}{x^2-9}$

c) $y = \dfrac{x^2-9}{x-3}$

d) $y = \dfrac{x^2+9}{x+3}$

Question 18 — 1 mark

Identify the value of $\ln e$.

a) 0

b) $\log e$

c) 1

d) e

Question 19 — 1 mark

When the polynomial function, $p(x)$, is divided by $(x-4)$ the remainder is 17. Identify which statement is true.

a) $p(4)=17$

b) $p(-4)=17$

c) $p(4)=0$

d) $p(-4)=0$

Question 20 — 1 mark

Identify the expression equivalent to $\frac{(n-6)!}{(n-4)!}$.

a) $(n-4)(n-5)$

b) $\frac{1}{(n-4)(n-5)}$

c) $(n-6)(n-5)$

d) $\frac{1}{(n-6)(n-5)}$

Question 21 1 mark

Identify the quadrant in which θ terminates if $\sec\theta = -\frac{4}{3}$ and $\sin\theta > 0$.

a) I

b) II

c) III

d) IV

Question 22 1 mark

Identify the expression that represents all angles that are coterminal with $\frac{\pi}{3}$.

a) $\frac{\pi}{3} + \pi k,\ k \in \mathbb{Z}$

b) $\frac{\pi}{3} + \pi k,\ k \in \mathbb{R}$

c) $\frac{\pi}{3} + 2\pi k,\ k \in \mathbb{Z}$

d) $\frac{\pi}{3} + 2\pi k,\ k \in \mathbb{R}$

Question 23 — 1 mark

115

Given $f(x)=\frac{1}{2}x-3$, state the coordinates of an invariant (unchanged) point when sketching the graph of $y=\sqrt{f(x)}$.

Question 24 2 marks

Solve, algebraically.

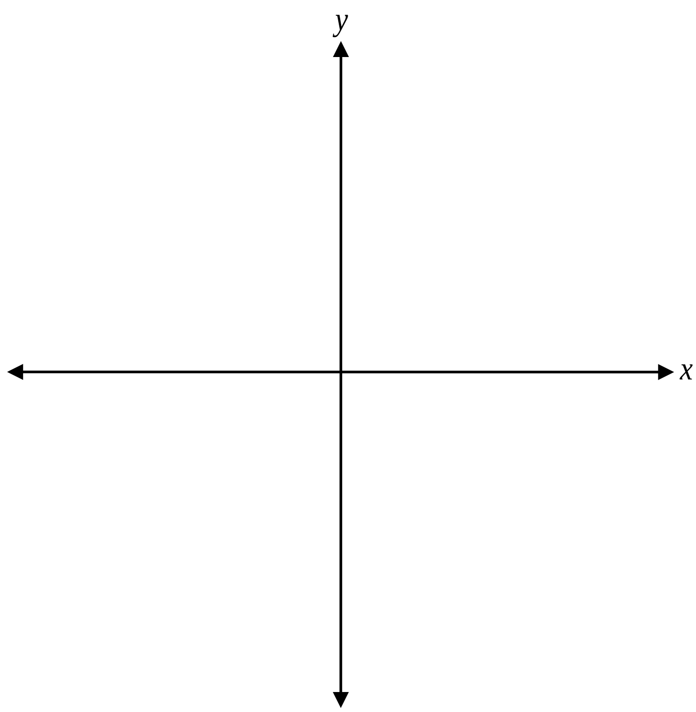
$$125^{3x+4} = \left(\frac{1}{5}\right)^{x}$$

Question 25 2 marks 117

Given $\sin\theta = -\frac{\sqrt{2}}{2}$ and $\cos\theta = -\frac{\sqrt{2}}{2}$, justify that the value of $\tan(2\theta)$ is undefined.

Question 26 3 marks

Sketch the graph of $y = -2\log_3 x$.



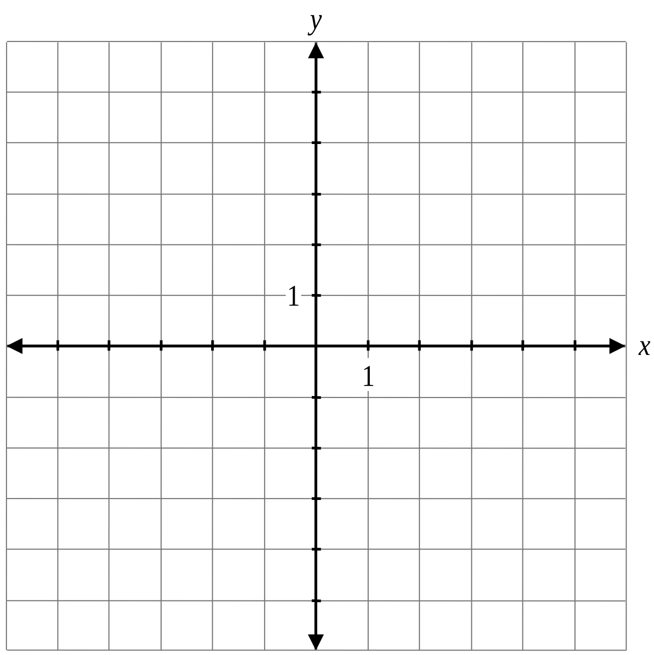
Question 27 2 marks 119

Evaluate.

$$\cos\left(\pi \bullet \sin\left(-\frac{\pi}{6}\right)\right)$$

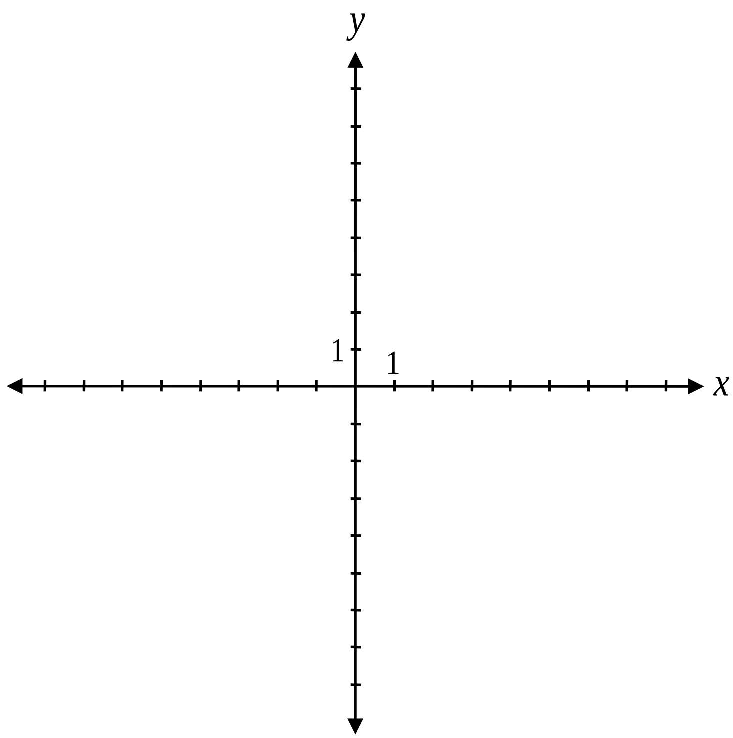
Question 28 a) 1 mark b) 1 mark 120 121

a) Given $f(x) = -x + 2,$ sketch the graph of $h(x) = f(f(x)).$



b) Explain why the domain of $h(x) = f(f(x))$ does not have any restrictions.

a) Sketch the graph of $f(x) = \dfrac{3x-5}{2x+4}$.



b) State the range of $f(x)$.

Range: ______________________________

Question 30 — 2 marks 124

Determine, algebraically, the equation of $p(x)$ that satisfies all of the following conditions:

- $p(x)$ is a polynomial function of degree 4
- $p(x)$ has a zero at 3 with a multiplicity of 2
- $p(x)$ has zeroes at -1 and -2
- $p(x)$ passes through the point $(2, 24)$.

$p(x) =$ ______________________________

Question 31 3 marks 125

Verify, by substitution, that the equation $\dfrac{\cos\theta + \cot\theta}{\cot\theta} = 1 + \sin\theta$ is true for $\theta = \dfrac{2\pi}{3}$.

Left-Hand Side	Right-Hand Side

Question 32 1 mark 126

Jyugo was asked to state the equation of the vertical asymptote(s) on the graph of $f(x)=\dfrac{3x-15}{x^2-5x}$.

His solution:

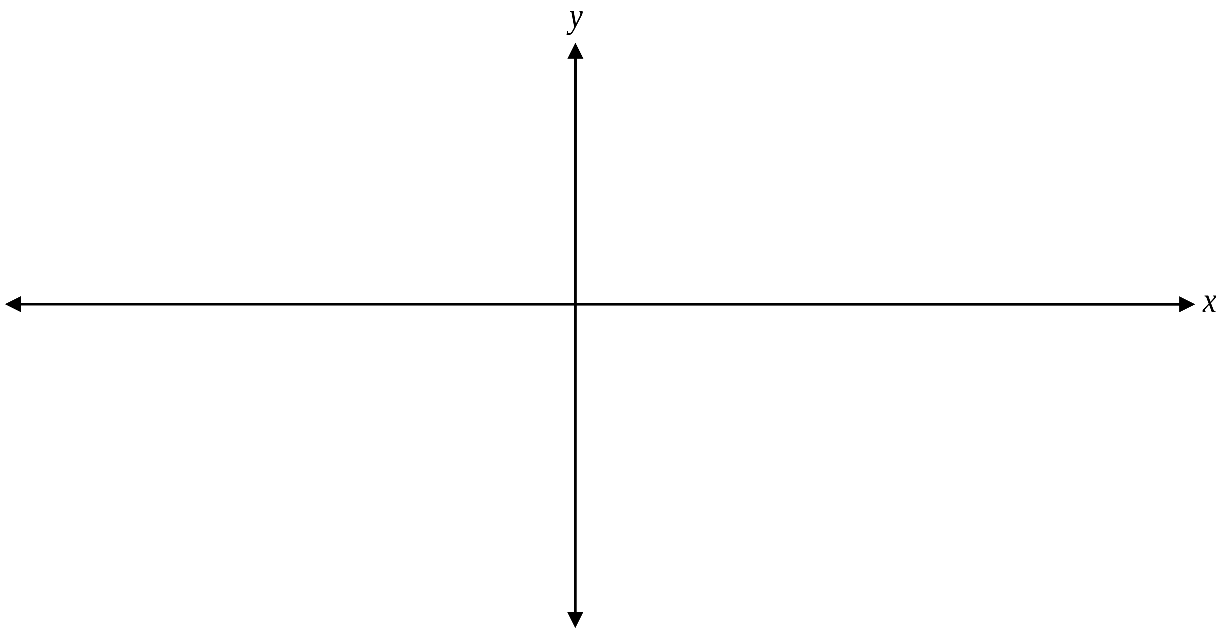
$$f(x)=\frac{3(x-5)}{x(x-5)}$$

$$x=0 \qquad x=5$$

Explain why his solution is incorrect.

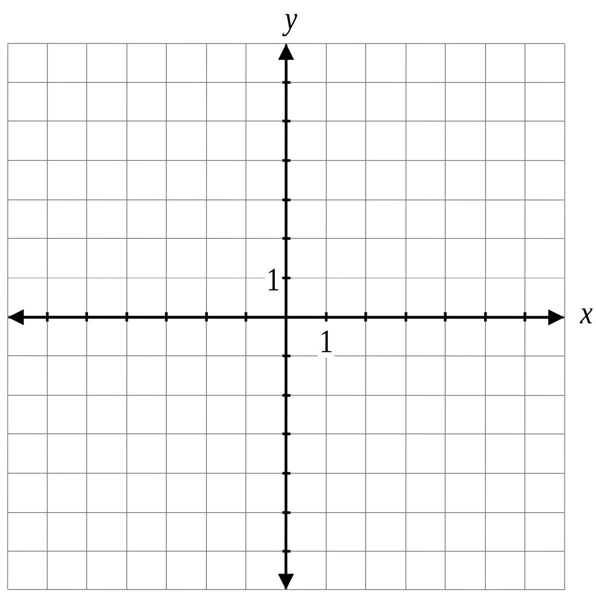
Question 33 3 marks 127

Sketch the graph of $y = \sin\left(\frac{1}{2}x\right) - 1$ over the interval $[-2\pi, 2\pi]$.



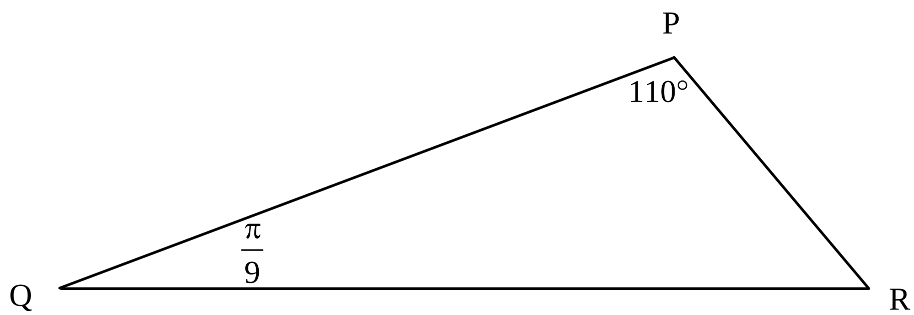
Question 34 4 marks 128

Sketch the graph of $y = \sqrt{-2x + 6}$.



Question 35 — 2 marks 129

Determine the measure of angle R, in radians.



Question 36 — 1 mark 130

Given $f(x) = \sqrt{x - 4}$, state the equation of the resulting function, $g(x)$, after a reflection over the x-axis.

$g(x) =$ ______________________

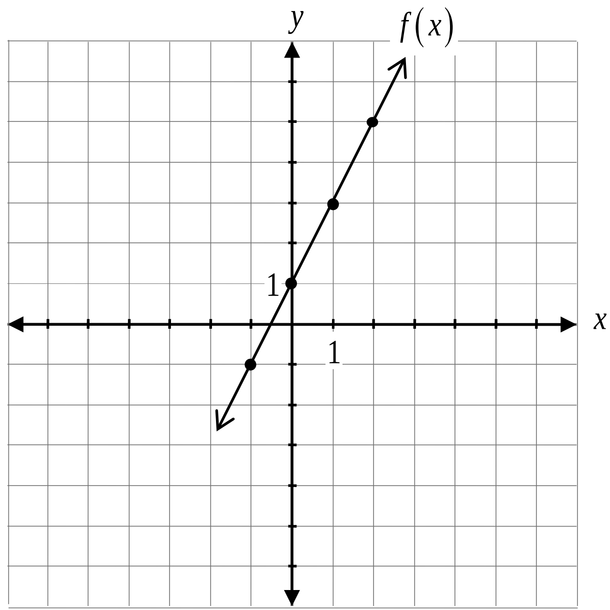
Question 37 a) 1 mark b) 3 marks 131 132

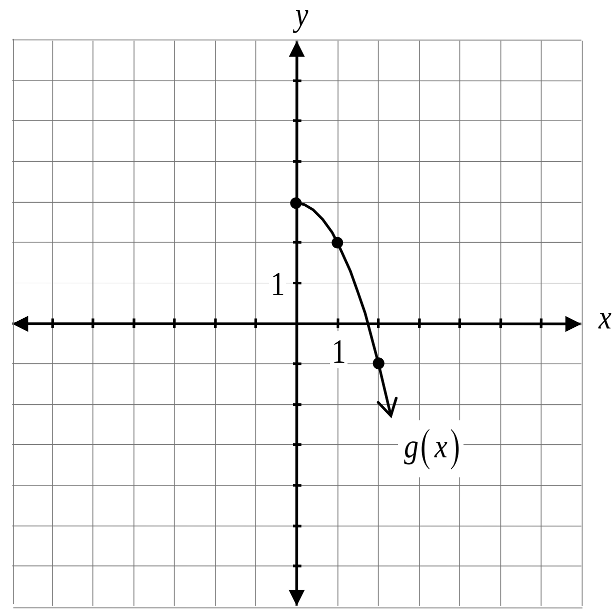
a) Determine the coterminal angle of $\frac{29\pi}{12}$ over the interval $[0, 2\pi]$.

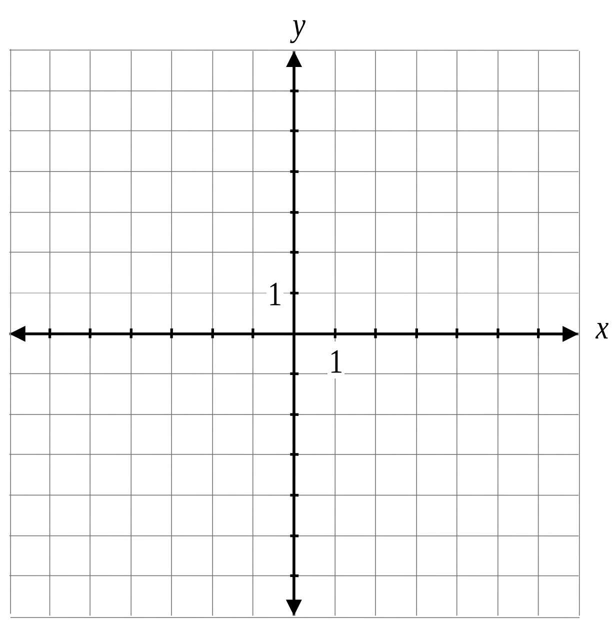
b) Determine the exact value of $\sin\left(\frac{29\pi}{12}\right)$.

Question 38 — 2 marks 133

Given the graphs of $f(x)$ and $g(x)$, sketch the graph of $h(x) = f(x) - g(x)$.







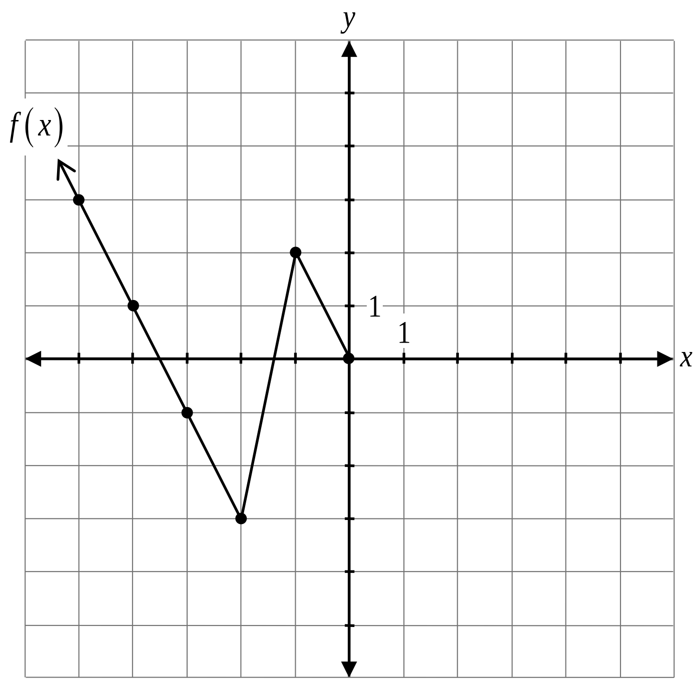
Question 39 3 marks

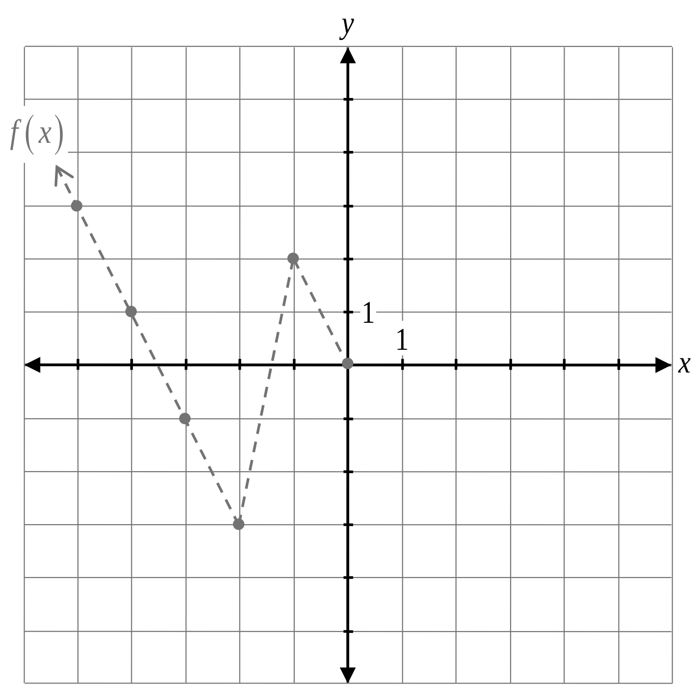
Determine an equation of a sinusoidal function that has the following characteristics:

- an amplitude of 3
- a period of 6
- a minimum value of -5

Question 40 — 1 mark

Given the graph of $y = f(x)$, sketch the graph after a reflection over the line $y = x$.





The graph of $f(x)$ has already been drawn for your reference.

No marks will be awarded for the graph of $f(x)$.

Question 41 a) 1 mark b) 1 mark 136 137

a) Determine the remainder when $3x^3+5x^2-13x-3$ is divided by $(x+3)$.

b) Is $(x+3)$ a factor of $3x^3+5x^2-13x-3$? Explain your reasoning.

Question 42 3 marks 138

State the range, the y-intercept, and the equation of the asymptote of the exponential function, $f(x) = 3^{x-1} + 2.$

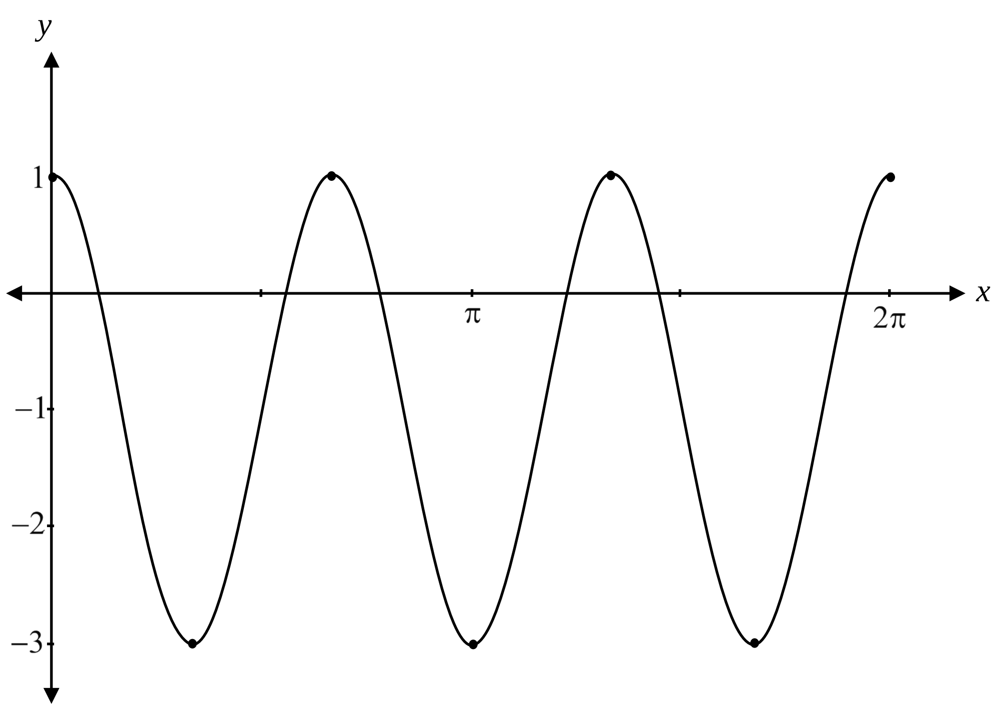
Range:______________________________

y-intercept:__________________________

Equation of the asymptote:______________

Question 43 1 mark 139

The graph of $y = 2\cos(3x) - 1$ below can be used to solve the equation $0 = 2\cos(3x) - 1$ over the interval $[0, 2\pi]$. Indicate on the graph where to find at least one solution to the equation $0 = 2\cos(3x) - 1.$



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